PHYSICS

Higher Level

Set B | Paper 2

Challenge mock examination

Time allowed: 150 minutes Maximum mark: 90

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

• A calculator and a clean current Physics data booklet are permitted throughout.

• All structured questions are compulsory. There are no optional topics or optional questions.

• Answer all questions. Show physical reasoning, working and units in written responses.

• Use the constants and assumptions given in a question. Otherwise use the data booklet and retain unrounded intermediate values.

• Give numerical answers to an appropriate number of significant figures. Diagrams are schematic unless explicitly stated to be to scale.

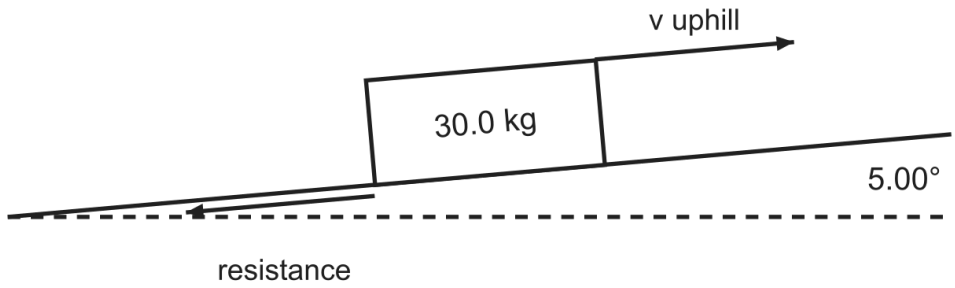
• The number in square brackets gives the maximum mark for a written part.

Thriving Scholars challenge practice | Current Physics syllabus | 2026

Paper 2 - Structured questions

2 Question 1 [10 marks]

A motorized inspection trolley of mass $30.0\,\text{kg}$ moves up a rail inclined at 5.00° to the horizontal. Rolling and aerodynamic resistance have resultant magnitude $F_r = bv^2$, opposite to the motion. The motor transfers 80.0% of its electrical input power to mechanical power. At a steady uphill speed of $3.00\,\text{m s}^{-1}$, input power is $300\,\text{W}$. Take $g = 9.81\,\text{m s}^{-2}$.



(a) Determine b and express its unit using fundamental SI units. [3]

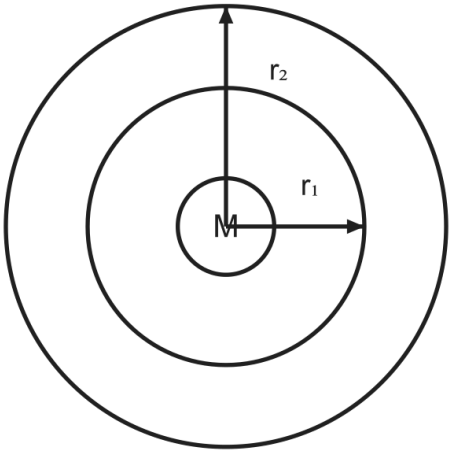
(b) Calculate the electrical input power required for a steady uphill speed of $6.00\,\text{m s}^{-1}$. [3]

(c) The trolley is now allowed to roll downhill with its motor disengaged. Determine its terminal speed under the same resistance model. [2]

(d) For very large uphill speeds, explain how the steady speed changes when electrical input power is multiplied by eight. State the approximation required. [2]

2 Question 2 [10 marks]

A probe of negligible mass moves in a circular orbit of radius $r_1 = 7.20 \times 10^7\,\text{m}$ around an isolated spherical planet, with orbital period $6.00\,\text{h}$. A second probe moves in a circular orbit of radius $r_2 = 1.60 r_1$. There is no significant atmospheric drag. Use $G = 6.67 \times 10^{-11}\,\text{N}\,\text{m}^2\text{kg}^{-2}$.



(a) Determine the planet mass from the first orbit. [3]

(b) Determine the second probe orbital period. [2]

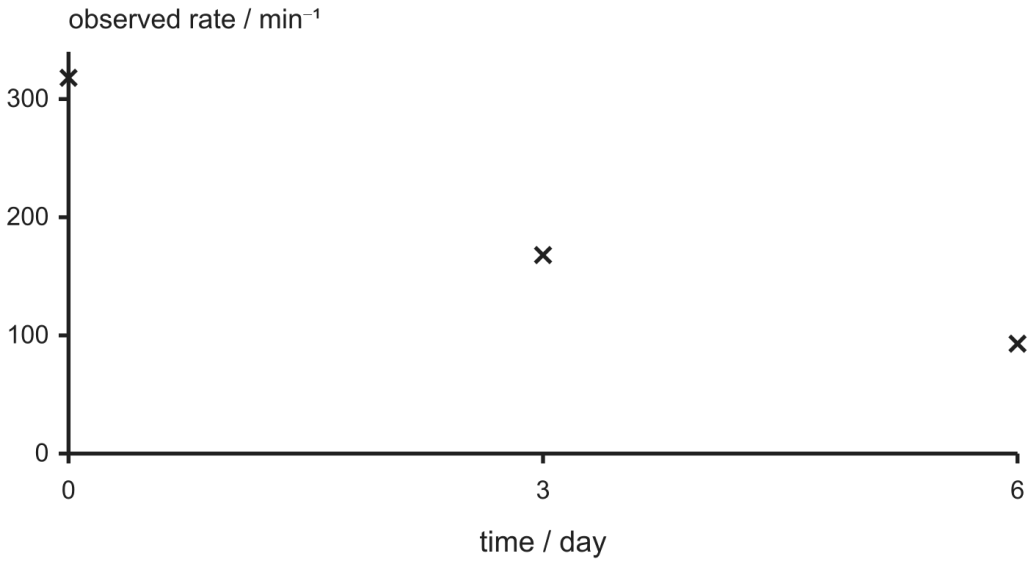
(c) Calculate $\phi(r_2) - \phi(r_1)$ and the magnitude of the mean radial field between these radii, defined by $\langle g \rangle = [\phi(r_2) - \phi(r_1)]/(r_2 - r_1)$. [3]

(d) Show that the local field magnitude equals this mean at radius $\sqrt{r_1 r_2}$, and determine that radius. [2]

2 Question 3 [10 marks]

An activated sample contains a parent nuclide P that decays to a stable daughter D. Initially no D is present. Laboratory model count rates at times 0, 3.00 and 6.00 days are 318, 168 and $93.0\,\text{min}^{-1}$. The observed rate is a single exponential plus a constant background. Detector calibration shows that 15.0% of parent decays are recorded, including all geometric and detection effects.

Each parent decay releases a total of $550\,\text{keV}$ as gamma radiation, either directly or in a two-photon cascade. Three possible photon energies are 210, 340 and $550\,\text{keV}$. Use $1\,\text{eV} = 1.602 \times 10^{-19}\,\text{J}$ and one day $= 86400\,\text{s}$.



(a) Determine the background rate and the parent half-life. Show how the equal time intervals constrain the background. [3]

(b) Determine the decay constant in s^{-1} and the initial number of parent nuclei. [2]

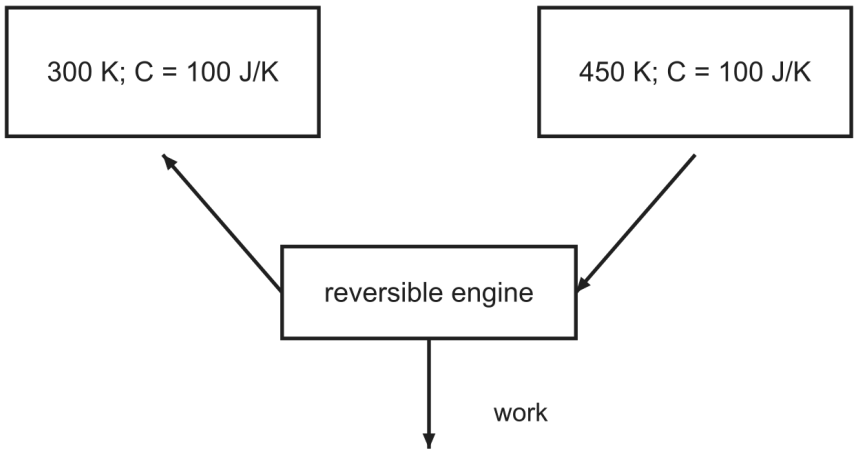
(c) Determine the number of daughter nuclei and the total emitted gamma energy after 9.00 days. [3]

(d) Draw a minimal energy-level diagram that can account for the three gamma energies. Explain why this is consistent with quantized nuclear energies. [2]

2 Question 4 | Question paper

2 Question 4 [10 marks]

Two solid blocks, each of constant heat capacity $C = 100\,\mathrm{J\,K^{-1}}$, start at $300\,\mathrm{K}$ and $450\,\mathrm{K}$. They are thermally insulated from their surroundings; useful work can be transferred to an external ideal work store in the second arrangement. No phase changes occur. For either block, use the supplied relation $\Delta S = C\ln(T_f/T_i)$.
In a second arrangement, a reversible engine operates between the two blocks, extracting work until their temperatures are equal. The engine returns to its initial state.



(a) Find the final temperature if the blocks are simply placed in direct thermal contact, with no work extracted. [2]

(b) Calculate their total entropy change in direct contact and explain its sign. [3]

(c) For the reversible-engine arrangement, determine the common final temperature and the maximum work extracted. [3]

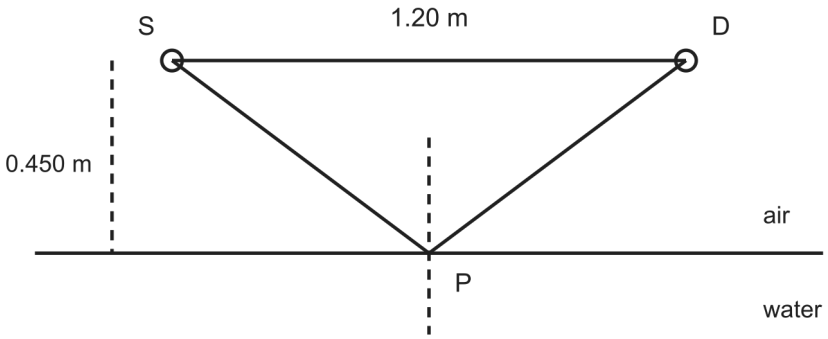
(d) A proposed engine would leave both blocks at $300\,\mathrm{K}$ and deliver the entire reduction in block internal energy as work, without any other change. Explain why energy conservation alone does not validate this proposal. [2]

2 Question 5 | Question paper

2 Question 5 [10 marks]

A small vehicle approaches a stationary vertical wall. Its onboard source emits $40.0\,\text{kHz}$ sound and its receiver detects a reflected frequency of $43.0\,\text{kHz}$. The speed of sound in air is $343\,\text{m}\,\text{s}^{-1}$. Treat reflection at the wall as preserving the frequency in the wall frame.

In a separate stationary arrangement, a sound source S and detector D are $1.20\,\text{m}$ apart, both $0.450\,\text{m}$ above a flat water surface. Sound reaches D directly and by reflection at P, midway horizontally between S and D. There is no phase change of the pressure wave at reflection. The sound speed in water is $1480\,\text{m}\,\text{s}^{-1}$, and both media are non-dispersive.



(a) Determine the speed of the approaching vehicle, accounting for both stages of the reflected Doppler shift. [3]

(b) For the stationary arrangement, determine the lowest positive frequency producing destructive interference of the direct and reflected pressure waves at D. [3]

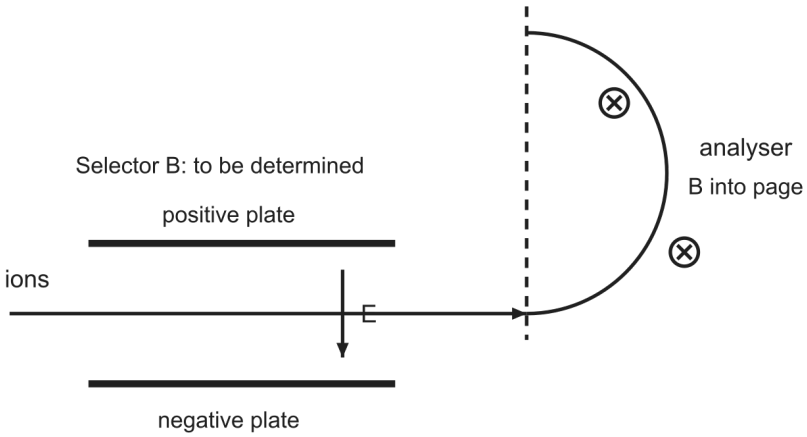
(c) Use wave refraction to decide whether a propagating transmitted ray in water exists for the ray SP at the same incident angle. Show your angular calculation. [3]

(d) Within the stated non-dispersive model, can changing only the frequency restore a transmitted ray at that incident angle? Explain. [1]

2 Question 6 [10 marks]

Singly positive ions are accelerated from rest through $4.20\,\text{kV}$. They enter crossed uniform fields: electric field $E = 2.40 \times 10^{4}\,\text{V m}^{-1}$ downwards and magnetic field of magnitude $0.0800\,\text{T}$. Selected ions travel horizontally to the right without deflection.

After the selector, they enter a region containing only a uniform magnetic field of $0.200\,\text{T}$ into the page. They follow a semicircle and hit a detector in the entry plane. Use $e = 1.602 \times 10^{-19}\,\text{C}$ and $1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$. Motion is non-relativistic.



Schematic: not to scale

(a) Determine the selected speed and the required direction of the selector magnetic field. [2]

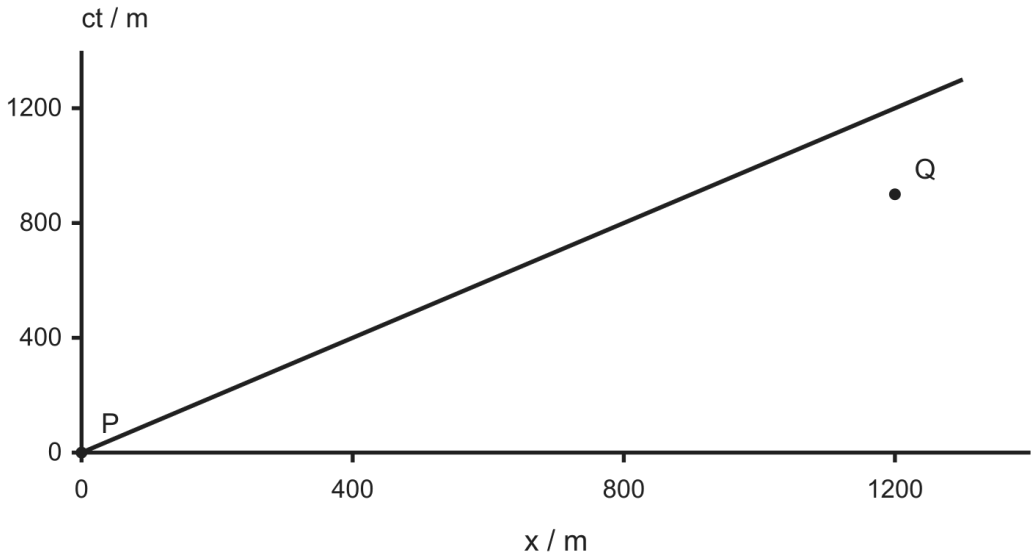
(b) Determine the selected ion mass and its approximate nucleon number, taking nuclear mass to be approximately $A\,\text{u}$. [3]

(c) Determine the separation of the detector impact point from the analyser entry point. [3]

(d) Ions of a heavier isotope with the same charge are accelerated through the same potential difference. Explain whether they pass the selector, and toward which plate they initially deflect. [2]

2 Question 7 [10 marks]

In inertial frame S, event P is a flash at $x = 0$, $ct = 0$. Event Q is a separately programmed flash at $x = 1200\,\text{m}$, $ct = 900\,\text{m}$. Frame S′ moves at $+0.800c$ relative to S and its origin coincides with P. Use $x' = \gamma(x - \beta ct), \qquad ct' = \gamma(ct - \beta x), \qquad \gamma = (1 - \beta^2)^{-1/2}$.
Use the convention $(\Delta s)^2 = (c\Delta t)^2 - (\Delta x)^2$.



(a) Calculate the invariant interval between P and Q and classify their separation. [2]

(b) Calculate the S′ coordinates of Q and state which flash occurs first in that frame. [3]

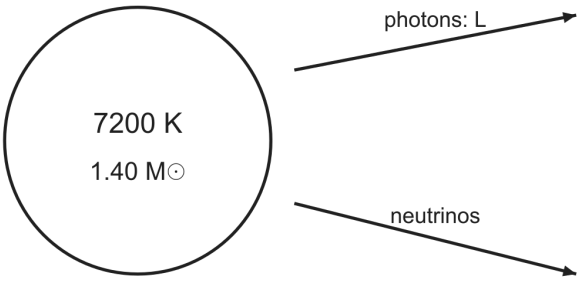
(c) Determine the speed of a frame in which the flashes are simultaneous and the spatial separation of the flashes in that frame. [2]

(d) Explain why the reversed time order creates no causal paradox, and why Q cannot have been triggered by a signal emitted at P. [3]

2 Question 8 [10 marks]

A main-sequence star is modelled as a blackbody with luminosity $8.00 \times 10^{26}\,\mathrm{W}$, effective surface temperature $7200\,\mathrm{K}$ and mass $1.40 M_{\odot}$, where $M_{\odot} = 2.00 \times 10^{30}\,\mathrm{kg}$. Its initial hydrogen mass fraction is 0.700. In this simplified lifetime model, 12.0% of that hydrogen mass can undergo fusion, converting 0.700% of the consumed hydrogen rest mass into energy. Neutrinos carry away 2.00% of the released energy; the rest supplies the stated luminosity.

For a reaction-rate estimate, use $26.7\,\mathrm{MeV}$ total energy released per completed four-proton fusion chain. Use $\sigma = 5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}K^{-4}}$, $c = 3.00 \times 10^{8}\,\mathrm{m\,s^{-1}}$, $1\,\mathrm{eV} = 1.602 \times 10^{-19}\,\mathrm{J}$ and one year $= 3.15576 \times 10^{7}\,\mathrm{s}$.



blackbody star model

(a) Determine the stellar radius. [2]

(b) Estimate the model main-sequence lifetime, assuming constant luminosity and the stated accessible fuel fraction. [3]

(c) Determine the number of completed fusion chains per second needed to supply the luminosity. [2]

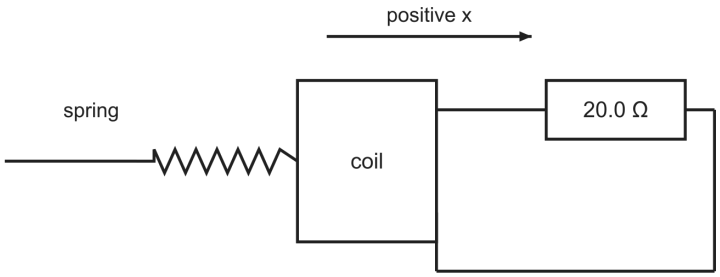
(d) Calculate the rate of stellar mass loss due to all the escaping radiation and neutrinos. Then explain how a small increase in core fusion rate can be moderated by hydrostatic feedback in a stable main-sequence star. [3]

2 Question 9 [10 marks]

A light rectangular coil attached to a horizontal spring oscillates along the x direction. The total moving mass is $0.0800\,\text{kg}$ and the spring constant is $12.0\,\text{N}\,\text{m}^{-1}$. The initial amplitude is $0.0400\,\text{m}$. Mechanical friction is negligible.

The coil has 200 turns, each of area $4.00 \times 10^{-4}\,\text{m}^2$. A static magnetic field is perpendicular to the coil, with constant spatial gradient $dB/dx = 8.00\,\text{T}\,\text{m}^{-1}$ across the operating region. Flux linkage is $NAB(x)$, where x is the coil centre. The circuit resistance is $20.0\,\Omega$; self-inductance is negligible.

For weak damping with force $F = -bv$, use the supplied amplitude envelope $a(t) = a_0 e^{-bt/(2m)}$.
Use the undamped period to estimate elapsed time for a given number of cycles.



B normal to the coil; dB/dx = 8.00 T/m

(a) Calculate the undamped period and initial maximum speed. [2]

(b) Calculate the initial maximum induced emf. [2]

(c) Using the rate of electrical energy dissipation and energy conservation, show that the magnetic damping coefficient is $b = (NA\,dB/dx)^2/R$. Calculate b and justify the force direction. [3]

(d) Estimate the fraction of initial oscillation energy remaining after ten cycles. Explain the limiting effect of disconnecting the external circuit. [3]

PHYSICS

Higher Level

Set B | Paper 2

Worked solution book

Complete the examination before consulting this book. Marking points indicate a method (M), accuracy (A) or reasoning (R) award. The number after the code gives its value. Award only points supported by the candidate's retained working.

Accept valid equivalent approaches and reasonable rounding. Apply error carried forward only where the subsequent method remains physically meaningful; do not reward a contradiction of stated conditions. A correct multiple-choice letter earns one mark; no negative marks apply.

MCQs are automatically marked on the portal. Written marks are self-assessed. Reports show these separately and do not calculate an official IB grade.

Paper 2 - Worked solutions

2 Question 1 [10 marks]

(a) [3 marks]

M1 Steady mechanical power is $0.800(300) = 240\ \text{W}$, so driving force is $240/3 = 80.0\ \text{N}$. Marking guidance: credit a valid subsequent method after an earlier arithmetic or graph-reading error; do not penalize the same numerical error twice. Require a correct physical model and units, and allow consistent use of rounded intermediate values.

M1 Balance gives $80.0 = 30.0(9.81)\sin 5^\circ + b(3.00)^2$.

A1 $b = 6.04\,\text{kg m}^{-1}$. Since $F = bv^2$, units are $(\text{kg m s}^{-2})/(\text{m}^2\text{s}^{-2}) = \text{kg m}^{-1}$.

(b) [3 marks]

M1 At $6.00\,\text{m s}^{-1}$, gravitational force is still $25.650\,\text{N}$, while resistance is $b(6.00)^2 = 217.400\,\text{N}$.

M1 Mechanical power is $(25.650 + 217.400)(6.00) = 1458.30\,\text{W}$; divide by efficiency.

A1 Input power is $1.82 \times 10^3\,\text{W}$ to three significant figures. It is not eight times $300\,\text{W}$ because the gravitational term scales only linearly with speed.

(c) [2 marks]

M1 At terminal speed, $bv^2 = mg\sin 5.00^\circ$.

A1 $v = \sqrt{25.650/6.03890} = 2.06\,\text{m s}^{-1}$ downhill.

(d) [2 marks]

R1 If $bv^2 \gg mg\sin 5^\circ$, the required mechanical power is approximately bv^3.

A1 With unchanged efficiency, $v \propto P_{\text{in}}^{1/3}$; multiplying power by eight approximately doubles steady speed.

2 Question 2 [10 marks]

(a) [3 marks]

M1 Equate gravitational and centripetal acceleration: $GM/r_1^2 = 4\pi^2 r_1/T_1^2$. Marking guidance: credit a valid subsequent method after an earlier arithmetic or graph-reading error; do not penalize the same numerical error twice. Require a correct physical model and units, and allow consistent use of rounded intermediate values.

M1 Use $T_1 = 21600\,\mathrm{s}$ in $M = 4\pi^2 r_1^3/(GT_1^2)$.

A1 $M = 4.74 \times 10^{26}\,\mathrm{kg}$.

(b) [2 marks]

M1 For the same central mass, $T_2/T_1 = (r_2/r_1)^{3/2} = 1.60^{3/2}$.

A1 $T_2 = 12.1\,\mathrm{h}$ to three significant figures.

(c) [3 marks]

M1 $\phi = -GM/r$, so $\Delta\phi = GM(1/r_1 - 1/r_2) > 0$.

A1 $\Delta\phi = 1.64 \times 10^8\,\mathrm{J\,kg^{-1}}$.

A1 $\langle g \rangle = \Delta\phi/(0.60 r_1) = 3.81\,\mathrm{m\,s^{-2}}$. The actual field direction is inward.

(d) [2 marks]

M1 $\langle g \rangle = GM[(r_2 - r_1)/(r_1 r_2)]/(r_2 - r_1) = GM/(r_1 r_2)$. Equating to GM/r^2 gives $r = \sqrt{r_1 r_2}$.

A1 $r = 9.11 \times 10^7\,\mathrm{m}$. It is the geometric, rather than arithmetic, mean radius.

2 Question 3 [10 marks]

(a) [3 marks]

M1 For background B, the common three-day decay factor requires $(168 - B)^2 = (318 - B)(93 - B)$. Marking guidance: credit a valid subsequent method after an earlier arithmetic or graph-reading error; do not penalize the same numerical error twice. Require a correct physical model and units, and allow consistent use of rounded intermediate values.

A1 This gives $B = 18.0\,\text{min}^{-1}$, so corrected rates are 300, 150 and $75.0\,\text{min}^{-1}$.

A1 The corrected rate halves in each 3.00-day interval; $t_{1/2} = 3.00$ days.

(b) [2 marks]

A1 $\lambda = \ln 2/(3.00 \times 86400) = 2.67 \times 10^{-6}\,\text{s}^{-1}$.

A1 $A_0 = (300/60)/0.150 = 33.333\,\text{s}^{-1}$ and $N_0 = A_0/\lambda = 1.25 \times 10^7$. Use unrounded λ in this calculation.

(c) [3 marks]

M1 After three half-lives, $N_P = N_0/8$ and $N_D = 7N_0/8$.

A1 $N_D = 1.09 \times 10^7$ daughter nuclei.

A1 Total gamma energy is $N_D(550 \times 10^3)(1.602 \times 10^{-19}) = 9.61 \times 10^{-7}\,\text{J}$. A two-photon cascade still carries total energy 550 keV per decay.

(d) [2 marks]

A1 Draw levels at 0, 340 and 550 keV, with downward arrows $550 \to 0$, $550 \to 340$ and $340 \to 0$. The two-step route emits $210 + 340 = 550$ keV. An intermediate level at 210 keV is an equivalent minimal arrangement.

R1 Discrete photon energies correspond to fixed differences between allowed nuclear energy levels, rather than a continuum of possible nuclear energies.

2 Question 4 [10 marks]

(a) [2 marks]

M1 No energy leaves, so $C(T_f - 300) = C(450 - T_f)$. Marking guidance: credit a valid subsequent method after an earlier arithmetic or graph-reading error; do not penalize the same numerical error twice. Require a correct physical model and units, and allow consistent use of rounded intermediate values.

A1 $T_f = 375\,\mathrm{K}$.

(b) [3 marks]

M1 $\Delta S = 100\ln(375/300) + 100\ln(375/450)$.

A1 $\Delta S = +4.08\,\mathrm{J\,K^{-1}}$.

R1 Direct heat transfer through a finite temperature difference is irreversible. The isolated system entropy increases, consistently with the second law.

(c) [3 marks]

M1 Reversibility requires zero total entropy change: $\ln(T_f/300) + \ln(T_f/450) = 0$, hence $T_f^2 = 300(450)$.

A1 $T_f = 367\,\mathrm{K}$ to three significant figures.

A1 $W_{\max} = C[300 + 450 - 2\sqrt{300(450)}] = 1.52 \times 10^3\,\mathrm{J}$. The reversible final temperature is below the direct-contact arithmetic mean.

(d) [2 marks]

R1 Both blocks at $300\,\mathrm{K}$ would give total entropy change $100\ln(300/450) = -40.5\,\mathrm{J\,K^{-1}}$.

R1 With no other change, the engine cannot compensate this entropy decrease. Although a first-law energy balance could be written, the proposal violates the second law and exceeds the reversible maximum work.

2 Question 5 [10 marks]

(a) [3 marks]

M1 The wall receives $f_w = fv/(v-u)$ because the source approaches it. Marking guidance: credit a valid subsequent method after an earlier arithmetic or graph-reading error; do not penalize the same numerical error twice. Require a correct physical model and units, and allow consistent use of rounded intermediate values.

M1 The moving receiver then detects $f_r = f_w(v+u)/v = f(v+u)/(v-u)$.

A1 $u = v(f_r - f)/(f_r + f) = 343(43.0 - 40.0)/(43.0 + 40.0) = 12.4\,\mathrm{m\,s^{-1}}$.

(b) [3 marks]

M1 The reflected length is $2\sqrt{0.600^2 + 0.450^2} = 1.50\,\mathrm{m}$; path difference is $0.300\,\mathrm{m}$.

M1 With no reflection phase reversal, the lowest positive destructive frequency has path difference $\lambda/2$, so $\lambda = 0.600\,\mathrm{m}$.

A1 $f = 343/0.600 = 572\,\mathrm{Hz}$ to three significant figures.

(c) [3 marks]

M1 Incident angle to the normal is $\theta_i = \tan^{-1}(0.600/0.450) = 53.1^\circ$.

M1 The critical angle is $\sin^{-1}(v_{\mathrm{air}}/v_{\mathrm{water}}) = \sin^{-1}(343/1480) = 13.4^\circ$.

A1 Since $53.1^\circ > 13.4^\circ$, the geometric-acoustics model gives total internal reflection and no propagating refracted ray. Equivalently the required refracted sine would exceed one.

(d) [1 mark]

R1 No. The critical angle depends on the speed ratio, which is frequency-independent under the stated non-dispersive assumption. Frequency changes the interference condition, not this transmission condition.

2 Question 6 [10 marks]

(a) [2 marks]

A1 $qE = qvB$ gives $v = E/B = 3.00 \times 10^5\,\mathrm{m\,s^{-1}}$. Marking guidance: credit a valid subsequent method after an earlier arithmetic or graph-reading error; do not penalize the same numerical error twice. Require a correct physical model and units, and allow consistent use of rounded intermediate values.

R1 The electric force is downward. The magnetic force on a positive ion moving right must be upward, requiring magnetic field into the page.

(b) [3 marks]

M1 $qV = mv^2/2$ gives $m = 2eV/v^2$.

A1 $m = 1.50 \times 10^{-26}\,\mathrm{kg}$.

A1 $m/\mathrm{u} = 9.007$, so the approximate nucleon number is $A = 9$.

(c) [3 marks]

M1 The magnetic force supplies centripetal force: $evB = mv^2/r$, so $r = mv/(eB)$.

A1 $r = 0.140\,\mathrm{m}$.

A1 The displacement after a semicircle in the entry plane is a diameter, $2r = 0.280\,\mathrm{m}$, above the entry point for the stated field direction.

(d) [2 marks]

R1 At fixed qV, speed is proportional to $m^{-1/2}$, so a heavier isotope is slower than the selected speed and is not transmitted undeflected.

R1 Its upward magnetic force qvB is smaller than the unchanged downward electric force qE; it initially deflects down toward the negative plate.

2 Question 7 [10 marks]

(a) [2 marks]

M1 $(\Delta s)^2 = 900^2 - 1200^2$. Marking guidance: credit a valid subsequent method after an earlier arithmetic or graph-reading error; do not penalize the same numerical error twice. Require a correct physical model and units, and allow consistent use of rounded intermediate values.

A1 $(\Delta s)^2 = -6.30 \times 10^5\,\mathrm{m}^2 < 0$, so the separation is spacelike.

(b) [3 marks]

M1 $\gamma = 1/\sqrt{1 - 0.800^2} = 5/3$.

A1 $x'_Q = (5/3)(1200 - 0.800 \times 900) = 800\,\mathrm{m}$.

A1 $ct'_Q = (5/3)(900 - 0.800 \times 1200) = -100\,\mathrm{m}$. Since P remains at $ct' = 0$, Q occurs first in S′.

(c) [2 marks]

A1 Simultaneity requires $900 - \beta(1200) = 0$, so the frame speed is $+0.750c$.

A1 With $\Delta ct'' = 0$, invariance gives $\Delta x'' = \sqrt{630000} = 794\,\mathrm{m}$. Direct Lorentz transformation gives the same value.

(d) [3 marks]

R1 In S, light travels only $900\,\mathrm{m}$ during the time between the events, less than their $1200\,\mathrm{m}$ separation.

A1 A signal from P to Q would require speed $\Delta x/\Delta t = (4/3)c > c$, so no allowed causal signal connects them.

R1 Spacelike events have no causal ordering enforced by a light-speed-limited influence. Different inertial observers may reverse their time order without reversing any actual cause and effect.

2 Question 8 [10 marks]

(a) [2 marks]

M1 Use $L = 4\pi R^2 \sigma T^4$. Marking guidance: credit a valid subsequent method after an earlier arithmetic or graph-reading error; do not penalize the same numerical error twice. Require a correct physical model and units, and allow consistent use of rounded intermediate values.

A1 $R = \sqrt{8.00 \times 10^{26} / [4\pi (5.67 \times 10^{-8})(7200)^4]} = 6.46 \times 10^8 \text{ m}$.

(b) [3 marks]

M1 Accessible hydrogen mass is $0.700(0.120)(1.40)(2.00 \times 10^{30}) = 2.352 \times 10^{29} \text{ kg}$.

M1 Energy available to the photon luminosity is $0.980(0.00700)(2.352 \times 10^{29})c^2 = 1.45212 \times 10^{44} \text{ J}$.

A1 $t = E/L = 1.81516 \times 10^{17} \text{ s} = 5.75 \times 10^9$ years. This estimate assumes constant luminosity, composition/fuel accessibility as specified and the given mass-conversion fraction.

(c) [2 marks]

M1 Photon energy available per completed chain is $0.980(26.7 \times 10^6)(1.602 \times 10^{-19}) \text{ J}$.

A1 Number of chains per second is $L/E_{\text{photon,chain}} = 1.91 \times 10^{38} \text{ s}^{-1}$. Do not treat the neutrino fraction as part of the observed photon luminosity.

(d) [3 marks]

A1 All escaping power is $L/0.980$, so $\dot{m} = L/(0.980c^2) = 9.07 \times 10^9 \text{ kg s}^{-1}$.

R1 An enhanced fusion rate heats the core and raises its pressure, tending to expand the star/core against gravity.

R1 Expansion cools the core and reduces the temperature-sensitive fusion rate, providing negative feedback around hydrostatic equilibrium. Accept an equivalent physically consistent account of pressure, gravity and temperature.

2 Question 9 [10 marks]

(a) [2 marks]

A1 $T = 2\pi\sqrt{m/k} = 0.513\,\text{s}$. Marking guidance: credit a valid subsequent method after an earlier arithmetic or graph-reading error; do not penalize the same numerical error twice. Require a correct physical model and units, and allow consistent use of rounded intermediate values.

A1 $\omega = \sqrt{k/m} = 12.247\,\text{rad}\,\text{s}^{-1}$ and $v_{\max} = \omega a_0 = 0.490\,\text{m}\,\text{s}^{-1}$.

(b) [2 marks]

M1 Faraday law gives $|\varepsilon| = |d(NAB)/dt| = NA|dB/dx|\,|v|$.

A1 $\varepsilon_{\max} = (200)(4.00\times10^{-4})(8.00)(0.489898) = 0.314\,\text{V}$.

(c) [3 marks]

M1 $P_R = \varepsilon^2/R = (NA\,dB/dx)^2v^2/R$. This energy is taken from the oscillator; $Fv = -P_R$, so $F = -[(NA\,dB/dx)^2/R]v$.

A1 $b = (0.640)^2/20.0 = 0.02048\,\text{kg}\,\text{s}^{-1} \approx 0.0205\,\text{kg}\,\text{s}^{-1}$.

R1 The force opposes velocity, as required by energy conservation and Lenz law. A force in the velocity direction would increase mechanical energy while also heating the resistor without a source.

(d) [3 marks]

M1 $E \propto a^2$, so $E(t)/E_0 = e^{-bt/m}$ and $t \approx 10T = 5.1302\,\text{s}$.

A1 $E/E_0 = e^{-(0.02048)(5.1302)/0.0800} = 0.269$. The small ratio $b/(2m\omega) \approx 0.0105$ supports the weak-damping period approximation.

R1 For an open circuit, $R \to \infty$, induced current and magnetic damping tend to zero. A changing flux can still produce an emf, but there is negligible Joule energy dissipation in this idealized limit.

End of worked solution book